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COVER FOR GAS DETECTOR

Abstract

A cover for a gas detector mounted on a ceiling surface includes a flat plate portion disposed along the ceiling surface and to which the gas detector is fixed, and a collar portion extending so as to surround the gas detector in the flat plate portion and inclined downward as the distance from the gas detector increases. At least one of the flat plate portion and the collar portion is provided with a plurality of through holes, and the plurality of through holes are arranged so as to surround the gas detector.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032698 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The technology disclosed herein relates to a cover for a gas detector.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2022-73236 (JP 2022-73236 A) discloses a fuel cell electric vehicle in which a housing space that houses a fuel cell is provided below a cabin of the vehicle. According to JP 2022-73236 A, the housing space is covered with a ceiling surface formed by the bottom of the cabin, and the ceiling surface is provided with a hydrogen sensor that detects a hydrogen concentration of the housing space.

SUMMARY

[0004] In a fuel cell electric vehicle or other equipment including a fuel cell, a portion where hydrogen may leak is not limited to the fuel cell. Hydrogen may leak from the fitting of a hydrogen tank, joints of various pipes, etc. Providing hydrogen detectors corresponding to the portions where hydrogen may leak causes an increase in cost. Therefore, there is a need for improvement to accurately detect hydrogen leakage by using as few hydrogen detectors as possible. Such a problem is not limited to the case of detecting hydrogen, but is also conceivable in various situations where gas is to be detected.

[0005] The present specification discloses a cover for a gas detector attached to a ceiling surface. The cover includes: [0006] a flat plate portion that is disposed along the ceiling surface and to which the gas detector is fixed; and [0007] a collar portion that extends to surround the gas detector on the flat plate portion and is inclined downward away from the gas detector.

[0008] In the above configuration, the cover includes the collar portion. Accordingly, when gas leakage occurs in a space including the ceiling surface, the gas drifting around the ceiling surface is easily retained in the vicinity of the gas detector by the action of the collar portion. As a result, the reliability of detection of the leaked gas by one gas detector can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0010] FIG. 1 is a view schematically showing a state in which a hydrogen detector is attached to a ceiling of a space;

[0011] FIG. 2 shows a cover according to a first embodiment from a lower perspective;

[0012] FIG. 3 is a cross-sectional view taken along III-III line of FIG. 2;

[0013] FIG. 4 shows a cover according to a second embodiment from a lower perspective;

[0014] FIG. 5 shows a cover according to a third embodiment from a lower perspective;

[0015] FIG. 6 is a cross-sectional view taken along VI-VI line of FIG. 5; and

[0016] FIG. 7 is a graph showing the change in hydrogen concentration with time.

DETAILED DESCRIPTION OF EMBODIMENTS

[0017] Embodiments will be described with reference to the drawings. Each of the drawings is merely an example, and the present embodiment is not limited to the illustrated contents. In

addition, since each of the drawings is an example, the illustrated shape is not accurate or a part thereof is omitted.

[0018] FIG. 1 schematically shows a state in which a hydrogen detector **20** capable of detecting hydrogen is attached to a ceiling surface **11** of a space **10**. The hydrogen detector **20** is an example of a “gas detector”. Hereinafter, the hydrogen detector **20** will be described as an example of the gas detector. However, a detector or a sensor that can detect a gas or a gas that is lighter than air and can be detected from the viewpoint of ensuring safety or the like corresponds to the gas detector assumed in the present embodiment.

[0019] The space **10** is a space in which a gas to be detected can leak. The ceiling surface **11** is not necessarily a horizontal surface, but is in any case a surface defining an area above the space **10**. The space **10** may be a closed space or a partially open space. As an example, the space **10** is a space secured under the floor of a passenger cabin of a fuel cell electric vehicle, and a fuel-cell, a hydrogen-tank, various pipes, and the like are disposed in the space **10**.

[0020] In FIG. 1, pipes **12** and **13** used for supplying the hydrogen gas **15** and a joint **14** connecting the pipe **12** and the pipe **13** are shown. The joint **14** is an example of a portion where the hydrogen gas **15** may leak. Although not shown in the drawings, it may be understood that the space **10** includes a plurality of locations that can be the source of such leakage. The space **10** is not limited to a space provided in a moving means such as a vehicle, and may be a space secured in various facilities such as a factory, an office, and a home, or a building.

[0021] According to FIG. 1, the hydrogen detector **20** is fixed to the cover **30**, and the cover **30** is attached to the ceiling surface **11**. That is, when the hydrogen detector **20** is said to be attached to the ceiling surface **11**, it includes both a case in which at least a part of the hydrogen detector **20** is directly attached to the ceiling surface **11** and a case in which it is indirectly attached to the ceiling surface **11** via the cover **30** or the like. The cover **30** may be referred to as a gas detector cover or the like. As a material for forming the cover **30**, various materials such as a metal and a resin are assumed. In consideration of an environment in which the hydrogen detector **20** is installed, durability required for the cover **30**, and the like, a suitable material is appropriately selected and the cover **30** is formed. Although not shown in the drawings, the hydrogen detector **20** is connected to cables buried in the ceiling or routed in the space **10** as necessary for communication and power supply securing.

[0022] The cover **30** according to the first embodiment will be described with reference to FIGS. 2 and 3.

FIG. 2 shows the cover **30** from a lower perspective. FIG. 3 shows a cross-sectional view through III-III of FIG. 2. According to the first embodiment, the cover **30** generally includes a flat plate portion **31** and a collar portion **32**. The flat plate portion **31** is disposed along the ceiling surface **11**. Further, the hydrogen detector **20** is fixed to the flat plate portion **31**. The term “along the ceiling surface **11**” does not refer to only a state parallel to the ceiling surface **11**. As described above, the ceiling surface **11** is not necessarily a horizontal surface, and may be a curved surface. Therefore, the flat plate portion **31** only needs to be substantially along the ceiling surface **11**, and may be inclined within a predetermined angle range with respect to the ceiling surface **11**, for example.

[0023] The flat plate portion **31** may be directly fixed to the ceiling surface **11**. However, according to the first embodiment, the flat plate portion **31** is fixed to the ceiling surface **11** via the support member **33** in consideration of workability such as screwing at the time of attachment. That is, the support member **33** is fixed to the ceiling surface **11**, and the flat plate portion **31** is fixed to the support member **33**. The support member **33** is also referred to as a bracket or the like. When the support member **33** is interposed between the ceiling surface **11** and the flat plate portion **31**, the flat plate portion **31** is fixed at a position separated from the ceiling surface **11** by a predetermined distance, for example, several centimeters downward. The shape of the support member **33** is not particularly limited. In addition, in the present embodiment, there is no particular limitation on a method of fixing the object and the object.

[0024] According to the first embodiment, the flat plate portion **31** has a square shape or a rectangular shape close to a square shape. The flat plate portion **31** may be a flat plate as a whole or may be a member including at least a flat plate-shaped portion. The hydrogen detector **20** includes a sensor unit **21** for detecting hydrogen, and a main body unit **22** that houses a circuit necessary for realizing the function, communication function, and the like of the sensor unit **21**. The hydrogen detector **20** is fixed to the lower surface **31a** of the flat plate portion **31** with the sensor unit **21** facing downward. In the flat plate portion **31**, a range in which the hydrogen detector **20** is fixed is at least a flat plate shape.

[0025] In the flat plate portion **31**, a collar portion **32** extends so as to surround the hydrogen detector **20**. The collar portion **32** inclines downward as it moves away from the hydrogen detector **20**. The collar portion **32** may be referred to as an eaves portion or the like. According to the first embodiment, the collar portion **32** extends from each of the four edges of the flat plate portion **31** and is connected as a whole, and surrounds the flat plate portion **31** and the hydrogen detector **20**. In addition, the shape in which the end portion (the distal end **32a**) of the collar portion **32** opposite to the end portion connected to the flat plate portion **31** is connected one round is a rectangular shape similar to the flat plate portion **31**, which is a square or a rectangular shape close to a square.

[0026] As can be seen from FIG. 2, the hydrogen detector **20** and the cover **30** are arranged substantially concentrically about the hydrogen detector **20**. Further, as can be seen from FIG. 3, the distal end **32a** of the collar portion **32** is located farther from the ceiling surface **11** than the hydrogen detector **20**. That is, the distal end **32a** is located below the sensor unit **21** of the hydrogen detector **20**.

[0027] Further, a plurality of through holes **34** are provided in at least one of the flat plate portion **31** and the collar portion **32**. According to the first embodiment, each through hole **34** is formed in a slit shape along each of the four sides of the flat plate portion **31**, and the plurality of through holes **34** are arranged so as to surround the hydrogen detector **20**. The through hole **34** may be simply referred to as a slit. In FIG. 2, in the portion where the flat plate portion **31** and the collar portion **32** are connected, the through hole **34** is formed on the collar portion **32** side, but the through hole **34** may be formed on the flat plate portion **31** side, or may be formed over the flat plate portion **31** and the collar portion **32**. Of course, the number of the through holes **34** is not limited to four as illustrated.

[0028] Further, a wall portion **35** protruding downward is provided on the lower surface **31a** of the flat plate portion **31**. The wall portion **35** extends to surround the hydrogen detector **20**. In FIG. 2, the wall portion **35** surrounds the hydrogen detector **20** without interruption, but the continuity of the wall portion **35** may be partially interrupted around the hydrogen detector **20**. According to FIG. 3, the wall portion **35** does not protrude downward from the distal end **32a** of the collar portion **32**. The wall portion **35** is provided between the hydrogen detector **20** and the through hole **34**. However, the wall portion **35** may be located outside the hydrogen detector **20** with respect to at least a part of the through holes **34**. Furthermore, the wall portion **35** may be provided on the lower surface of the collar portion **32**.

[0029] The cover **30** according to the second embodiment will be described with reference to FIG. 4. FIG. 4 shows the cover **30** from a lower perspective, similar to FIG. 2. Only differences from the first embodiment will be described with respect to the second embodiment. A cross-sectional view of the second embodiment may be referred to FIG. 3 as in the first embodiment. As shown in FIG. 4, according to the second embodiment, the flat plate portion **31** and the collar portion **32** extending from the edge of the flat plate portion **31** have a circular shape. That is, the shape of the cover **30** when viewed from below may be circular.

[0030] In the second embodiment, each of the plurality of through holes **34** is also a slit curved along the curve of the edge of the flat plate portion **31**, the circular formed by connecting the plurality of through holes **34**, like the flat plate portion **31** and the collar portion **32**, the hydrogen detector **20** It is concentric around. Further, according to FIG. 4, the wall portion **35** surrounding

the hydrogen detector **20** is also concentric around the hydrogen detector **20**.

[0031] The cover **30** according to the third embodiment will be described with reference to FIGS. 5 and 6. FIG. 5 shows the cover **30** from a lower perspective, similar to FIGS. 2 and 4. FIG. 6 shows a cross-sectional view through VI-VI of FIG. 5. Also in the third embodiment, only the difference from the first embodiment will be described. According to the third embodiment, a through-hole **31b** is formed in the center of the flat plate portion **31** so as not to allow the main body unit **22** of the hydrogen detector **20** to pass therethrough. The main body unit **22** is fixed on the flat plate portion **31**, that is, in a space between the flat plate portion **31** and the ceiling surface **11**.

[0032] The through-hole **31b** is formed so as to allow the sensor unit **21** of the hydrogen detector **20** to pass therethrough, and the sensor unit **21** protrudes downward from the lower surface **31a** via the through-hole **31b** and faces downward. The support member **33** is shaped so as not to interfere with the main body unit **22** accommodated in the space between the flat plate portion **31** and the ceiling surface **11**. Alternatively, the support member **33** and the main body unit **22** may be partially in contact with or connected to each other.

[0033] In this third embodiment, the position of the hydrogen detector **20** is closer to the ceiling surface **11** than in the first embodiment or the second embodiment. It can be said that the hydrogen detector **20** is advantageous to detect hydrogen by the amount close to the ceiling surface **11**. Needless to say, in the third embodiment, the shape of the cover **30** may be a circular shape as shown in the second embodiment.

[0034] As described above, according to the present embodiment, the cover **30** for the gas detector (for example, the hydrogen detector **20**) attached to the ceiling surface **11** includes the flat plate portion **31** disposed along the ceiling surface **11** and to which the gas detector is fixed, and the collar portion **32** extending so as to surround the gas detector in the flat plate portion **31** and inclining downward as the distance from the gas detector increases. According to the above configuration, when the leakage of the hydrogen gas **15** occurs in the space **10** as shown in FIG. 1, the hydrogen gas **15** drifting around the ceiling surface **11** can be held in the vicinity of the hydrogen detector **20** by the action of the collar portion **32** of the cover **30**. As a result, the hydrogen concentration in the vicinity of the hydrogen detector **20** is stabilized, and the reliability of detection by one hydrogen detector **20** with respect to the leaked hydrogen gas **15** can be improved.

[0035] Further, according to the present embodiment, a plurality of through holes **34** may be provided in at least one of the flat plate portion **31** and the collar portion **32**, and the plurality of through holes **34** may be arranged so as to surround the gas detector. According to the above configuration, even if the installation position of the hydrogen detector **20** is a position deviated from directly above the leakage position (for example, the joint **14**) of the hydrogen gas **15**, the hydrogen detector is ejected from the leakage position and then moves along the ceiling surface **11**. The diffusing hydrogen gas **15** can enter the inside of the cover **30** from various directions through the respective through holes **34**. As a result, the hydrogen concentration inside the cover **30**, that is, in the vicinity of the hydrogen detector **20** is stabilized, and the reliability of the detection by the hydrogen detector **20** can be further increased.

[0036] Further, according to the present embodiment, the lower surface of the flat plate portion **31** may be provided with a wall portion **35** that protrudes downward and extends so as to surround the gas detector.

According to the above configuration, the effect of stabilizing the hydrogen concentration in the vicinity of the hydrogen detector **20** by holding the hydrogen gas **15** in the vicinity of the hydrogen detector **20** by the action of the wall portion **35** surrounding the periphery of the hydrogen detector **20** can be further enhanced.

[0037] Incidentally, the inclination angle θ (see FIG. 3) of the collar portion **32** with respect to the flat plate portion **31** may be set to, for example, 30 degrees to 60 degrees. If the inclination angle θ is too small, the effect of causing the hydrogen gas **15** to stay in the vicinity of the hydrogen

detector **20** by the cover **30** is small. On the other hand, if the inclination angle θ is too large, the hydrogen gas **15** leaked at a position not directly under the hydrogen detector **20** cannot be efficiently guided to the inside of the cover **30**. Therefore, a value range of 30 degrees to 60 degrees can be said to be an example of a value range suitable as the inclination angle θ .

[0038] Further, according to the present embodiment, the distal end **32a** of the collar portion **32** may be located farther from the ceiling surface **11** than the gas detector. According to the above configuration, the distal end **32a** of the collar portion **32** is located below the hydrogen detector **20**. Therefore, it is possible to suppress the hydrogen gas **15** staying in the vicinity of the hydrogen detector **20** from flowing out of the cover **30** by the collar portion **32**.

[0039] It is assumed that the hydrogen detector **20** is capable of detecting hydrogen having a predetermined concentration, for example, a concentration of 4% or more. When the hydrogen gas **15** leaked into the space **10** moves along the ceiling surface **11** after being raised, the hydrogen gas is diffused while meandering up, down, left, and right. Further, the hydrogen concentration decreases as the distance from the ceiling surface **11** decreases.

[0040] FIG. 7 graphically illustrates the change in hydrogen concentration over time in the vicinity of the hydrogen detector **20** at a predetermined position that is not directly above the leak position of the hydrogen gas **15**, but a predetermined number of centimeters below the ceiling surface **11**. In FIG. 7, the solid line graph shows the hydrogen concentration in the vicinity of the hydrogen detector **20** having the cover **30** of the first embodiment located at the predetermined position. On the other hand, in FIG. 7, a graph indicated by a two-dot chain line indicates the hydrogen concentration in the vicinity of the hydrogen detector **20** that is located at the predetermined position and does not include the cover **30**. Note that the hydrogen concentration shown in FIG. 7 can be interpreted as a concentration obtained by using a predetermined detector having a higher detection capability than the hydrogen detector **20**, which is different from the hydrogen detector **20**, for experiment.

[0041] As can be seen from FIG. 7, according to the configuration without the cover **30**, the variation in the hydrogen concentration is large and is less than 4% at many timings. On the other hand, according to the configuration having the cover **30**, the hydrogen concentration is substantially stable at 4% or more. Therefore, in the conventional configuration without the cover **30**, even if the leakage of the hydrogen gas **15** occurs, the hydrogen concentration is not stabilized in the vicinity of the hydrogen detector **20** that is not located directly above the leakage position, and the hydrogen detector **20** may not be able to detect hydrogen. On the other hand, according to the present embodiment, the hydrogen concentration in the vicinity of the hydrogen detector **20** is easily stabilized at a predetermined concentration or more by the action of the cover **30**. Therefore, a situation in which the hydrogen detector **20** cannot detect hydrogen even though the hydrogen gas **15** leaks at any position in the space **10** can be eliminated as much as possible. In this way, in the present embodiment, since the hydrogen detector **20** is highly reliable in detecting the leakage of hydrogen, it is possible to carry out a wide range of hydrogen detection in one hydrogen detector **20**, it is possible to contribute to cost reduction by reducing the number of necessary hydrogen detectors **20**.

[0042] The shape of the cover **30** will be described. According to the drawings, the collar portion **32** extends from the edge of the flat plate portion **31** toward the side away from the hydrogen detector **20** and downward. However, the collar portion **32** may extend from a position closer to the hydrogen detector **20** than the edge of the flat plate portion **31** in the flat plate portion **31**.

[0043] Further, a position where the collar portion **32** is connected to the flat plate portion **31** and a peripheral edge of the hydrogen detector **20** may be close to each other so that there is almost no distance. That is, the flat plate portion **31** may have a size equivalent to that of the hydrogen detector **20** as the minimum size required for fixing the hydrogen detector **20**, and the collar portion **32** may be connected to the peripheral edge of the flat plate portion **31** having such a size.

The shape of the cover **30** when viewed from below is not limited to the shape shown in the

drawing, and various shapes such as an elliptical shape and a polygon other than a quadrangle can be adopted.

[0044] While specific examples of the technology disclosed herein have been described in detail above, these are merely illustrative and do not limit the scope of the claims. Various modifications and variations of the specific examples described above are included in the technology described in the claims. In addition, the technical elements described in the present specification or the drawings exhibit technical usefulness alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. Further, the technology illustrated in the present specification or the drawings achieves a plurality of objects at the same time, and has technical usefulness by achieving one of the objects.

Claims

1. A cover for a gas detector attached to a ceiling surface, the cover comprising: a flat plate portion that is disposed along the ceiling surface and to which the gas detector is fixed; and a collar portion that extends to surround the gas detector on the flat plate portion and is inclined downward away from the gas detector.
 2. The cover according to claim 1, wherein: at least either of the flat plate portion and the collar portion has a plurality of through holes; and the through holes are arranged to surround the gas detector.
 3. The cover according to claim 1, wherein a lower surface of the flat plate portion includes a wall portion that protrudes downward and extends to surround the gas detector.
 4. The cover according to claim 1, wherein an inclination angle of the collar portion with respect to the flat plate portion is 30 degrees to 60 degrees.
 5. The cover according to claim 1, wherein a tip end of the collar portion is positioned farther from the ceiling surface than the gas detector.
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